Lab 6: Radiometer

Goal: Measure rotation speed at five lamp gaps from 1.0 to 4.0 cm, the coast-down after the lamp is switched off, and the glass temperature before and after each run. Compare speed with measured lux and check whether it changes during a run.

A. Prepare your bulb

1. Record your **bulb number** from the label on its base. Use the same bulb, lamp and lamp power for every run. Bulbs differ in internal pressure and friction, so results from different bulbs are not interchangeable.
2. Measure **room temperature** and the **glass temperature** at the marked point on the bulb equator facing the lamp. Use the same point and thermometer for every reading; if the infrared thermometer has an emissivity setting, use 0.95. The thermometer reads the **glass, not the vanes**: infrared thermometers cannot see through glass.
3. A **cooled start** means the glass is within 1 °C of room temperature and the rotor is at rest.
4. **No rotation is a result.** If the rotor does not start turning within 30 s at a gap, record “no rotation” for that gap, skip its movie, and leave that row out of the comparison log. The largest gap that still turns the rotor indicates the start-up friction.

B. Set the gaps and measure light

5. Set the gap between the **light-emitting front surface of the lamp** and the **nearest outer glass surface**, at rotor height, with spacer blocks or a ruler. Remove any spacer before switching on. Keep the lamp and glass from touching; do not substitute a hot incandescent or halogen lamp.

Recording	Lamp-to-glass gap	Movie name
First test	1.0 cm	1cm_1.mp4
Second test	1.5 cm	1p5cm.mp4
Third test	2.0 cm	2cm.mp4
Fourth test	3.0 cm	3cm.mp4
Fifth test	4.0 cm	4cm.mp4
Repeat after cooling	1.0 cm	1cm_repeat.mp4

6. **Measure lux at the rotor position.** For each gap, lift the bulb off its mark, place the lux meter sensor where the rotor centre was, facing the lamp, and read it. Return the bulb to the same mark. If a W/m² light meter is available, record that reading too.
7. Place the black background **25 cm behind the rear glass surface**, outside the strongest direct beam. Keep it and the camera fixed.
8. Once, at the 1.0 cm gap after a run, measure the temperature of the lamp’s front surface. A warm lamp face also heats the glass.

C. Camera settings

Resolution: 1280 × 1024 pixels. Capture frame rate: 1000 frames/s. The remaining camera settings are preset at the station.

D. Record each run

9. Record a short test. In Playback, check that a selected vane passage is clear. Watch one full turn and count **p = selected passage events per revolution**. If following one identifiable vane, p = 1; do not assume p = 4.
10. Read the glass temperature. Start from a cooled bulb with the rotor at rest.
11. Start recording and switch on the lamp. Capture at least **seven consecutive passage marks** (six intervals).
12. **Switch the lamp off and keep recording** for the rest of the available recording time, at least 10 s. Note whether the rotor slows smoothly, stops or turns backward.
13. Immediately read the glass temperature again.
14. Save the useful segment as **H.264, Text Overlay off**, with the movie name from the table. Note the **light-on frame** and the **light-off frame** in Playback.
15. Wait until the glass is within 1 °C of room temperature before the next run. Move only the lamp. Record the 1.0 cm repeat last. If a run starts warm, record that fact; do not call it a cooled repeat.

Use real recording time: at 1000 fps, 10,000 frames represent 10 s, regardless of slow-motion playback length. If the recording cannot hold the light-on passages and 10 s after light-off, extend the station recording duration before proceeding.

> **Keep for each movie:** bulb number, gap, lux, capture fps, p, light-on frame, light-off frame, room temperature, and glass temperature at start and end. Enter them in **MIE402_Lab6_Run_Log.csv**. Keep the original movie.

E. Mark passages and calculate rpm

16. Download and unzip **MATLAB files > Complete bundle** from the Lab 6 page. Run **Mark_Lab6_Passage_Frames.m** and select a movie. Left/Right navigate; Up/Down change the step; Space marks; Backspace undoes; Q saves.
17. Use a one-frame step at each crossing. Mark the **same passage event in the same direction**, in increasing frame order. Mark the light-on part first; the program saves a file ending **_passages.csv**.
18. Run **Run_Lab6_Radiometer.m**, select the passage CSV, and enter **1000 fps** and your verified p. If the actual capture rate differs, enter that rate. Check unexpected intervals against the movie.

One-interval check: elapsed seconds = frame-number difference / capture fps; rpm = 60 / (p × elapsed seconds). Seven marks give six intervals.

F. Coast-down time constant

19. For the first 1.0 cm run, mark the passages **after the light-off frame** in a second file, saved as **1cm_1_coast_passages.csv**. Run Run_Lab6_Radiometer.m on it.
20. Open the exported **interval_speed.csv**. Plot ln(rpm) against time. With the lamp off and a linear loss, $I\,d\omega/dt = -b\omega$, so $\omega = \omega_0 e^{-t/T}$ and the slope is $-1/T$, where $T = I/b$. Ignore the first 1–2 s, when the vanes are still warm. Enter T as **coast_time_constant_s**.
21. A straight line supports a linear loss; a line that curves down at low speed suggests constant pivot friction. The tool reports speed, not direction: state from the movie whether the rotor reversed.

G. Compare the gaps

22. Open **Lab6_Comparison_Log.csv**. The six filenames and gaps are already entered; delete any row recorded as no rotation. Fill lux, p, light-on frame, initial state, bulb number, temperatures, light-off frame and T. Keep capture_fps = 1000 unless the actual camera rate differed.
23. Choose the **same physical start/end times after light-on** that all compared movies cover, before light-off, with enough passage marks. Enter them in window_start_s and window_end_s for every row. Time after light-on = (frame − light-on frame) / capture fps.
24. Run **Compare_Lab6_Illumination_Runs.m** and select the completed log. Review rpm versus time and mean rpm versus measured lux. Compare the first and repeated 1.0 cm tests, and early versus late speed.

H. Put these results in the report

1. One labeled setup photo, your bulb number, and a table of gap, lux, glass temperature at start and end, and mean rpm.
2. Cumulative turns and rpm versus time for one recording, including light-off.
3. Mean rpm versus lux, with the repeated 1.0 cm test identified.
4. The coast-down plot and your value of T, with one sentence on what it shows about losses.
5. A short explanation of early/late changes, glass heating and repeatability, using your Q6 molecular Monte Carlo result to discuss temperature contrast and gas momentum transfer.

Use only relevant theory. Lux is not absorbed radiant power, the glass temperature is not the vane temperature, and the simple molecular wall model does not directly predict the bulb rpm.

Optional: filters at a fixed gap

If filter sheets are available, keep the lamp at 1.0 cm and place one sheet flat against the lamp face. **Neutral-density** sheets (0.3, 0.6, 0.9) reduce the light by about 2, 4 and 8 times without changing its colour, so speed can be compared with light while the lamp's own warmth stays the same. A **coloured** sheet changes lux far more than it changes the absorbed power: compare rpm per lux with and without it. Measure lux behind the sheet and label these runs in the run log.

Before leaving

Check that all movies, passage CSVs (including the coast-down file), the completed run log and comparison log, and MATLAB figures are saved and shared with your group. Complete Q6 in Pre-Lab 6 before your section. **Lab 6 report: December 7, 11:59 PM.**